

COMP 0085 Approximate Inference.

- Factor model: $P(X) = \frac{1}{Z} \prod_i f_i(X_{C_i})$ J indexes factors, G contains indices of variables adjacent to factors.
- Conditional independence: $X \perp\!\!\!\perp Y | V$ if every path between X and Y contains some $V \in V$.
 $\Rightarrow \exists$ a factorisation $P(X, Y, V, \dots) = \frac{1}{Z} g_X(X, V_X, Q_X) g_Y(Y, V_Y, Q_Y) g_R(Q_R, V_R)$.
- Variables are neighbours if they share a common factor.
- $X \perp\!\!\!\perp Y | ne(X)$, $\forall Y \notin \{X \cup ne(X)\}$ X is conditionally independent of all non-neighbours given its neighbours.
- $ne(X)$ is a Markov blanket for X and in fact it is minimal such set = Markov boundary.
- Undirected graphical models: Markov networks. Neighbours in a Markov net share a factor. non-neighbours cannot share a factor.
Joint probability factors over the maximal cliques G of the graph.
- Cliques are fully connected subgraphs (every two nodes are connected). Maximal cliques are cliques not contained in other cliques (cannot be enlarged to add more vertex).
- Directed Acyclic graphical (DAG) model: $P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | X_{pa(i)})$ $pa(i)$ are parents of node i . (Bayesian networks)
- Bayes-ball algorithm: Can you get a ball from X to Y without being blocked by V ? If so, $X \not\perp\!\!\!\perp Y | V$.
 - Nodes $V \notin V$ pass balls down or up chains: $\rightarrow V \rightarrow$ or $\leftarrow V \leftarrow$
 - Nodes $V \notin V$, bounce balls from children to children, via parent V
 - Nodes $V \in V$, bounce balls from parents to parents (including returning the ball whence it came) via children V .
- Markov boundary for X in DAG: $\{pa(X) \cup ch(X) \cup pa(ch(X))\}$
- Each graph G implies conditional independence statements $C(G) = \{X_i \perp\!\!\!\perp Y_i | V_i\}$
For each set C , we can define a family of distributions that satisfy all statements in C .
 $P_{CC(G)} = \{P(X) : P(X_i, Y_i | V_i) = P(X_i | V_i) P(Y_i | V_i) \text{ for } \forall X_i \perp\!\!\!\perp Y_i | V_i \text{ in } C\}$ Independence view
 G may also encode a family of distributions by functional form
 $P_G = \{P(X) : P(X) = \frac{1}{Z} \prod_i f_i(X_{C_i}), \text{ for some non-negative functions } f_i\}$ factorisation view.
 - For directed graphs, $P_G = P_{CC(G)}$ $P(X) = \prod_i P(X_i | X_{pa(i)})$ global Markov $\Leftrightarrow$ factorisation.
 - For undirected graphs, $P_G = P_{CC(G)}$ if all distributions are positive, i.e. $P(X) > 0 \forall X$ (Hammersley-Clifford Theorem) $P(X) = \frac{1}{Z} \prod_i f_i(X_{C_i})$ by cliques
 - For factor graphs, there are some cases $P_G \neq P_{CC(G)}$
 - Expressive advantage: Every undirected model can be written as a factor graph, but not every factor graph collapses to an undirected graph with same family.
- Adding edges to a graph $\Rightarrow$ removing conditional independency statements $\Rightarrow$ conditional independence set shrinks $\Rightarrow$ family of distribution enlarges. Denser graphs being more expressive but less constrained in conditional independence.
- polytrees are tree-structured DAGs that may have more than one root.
 $P(X) = \prod_i P(X_i | X_{pa(i)}) = \prod_i f_i(X_{C_i})$ $C_i = i \cup pa(i)$ and $f_i(X_{C_i}) = P(X_i | X_{pa(i)})$.
- Undirected tree, all maximal cliques are of size 2, factor graph has pairwise factors.
 $P(X) = \frac{1}{Z} \prod_{(i,j) \in E} f_{ij}(X_i, X_j)$.
- Single rooted directed tree can be written as a product of pairwise factors $\Rightarrow$ undirected tree.
 $P(X) = P(X_r) \prod_{i \neq r} P(X_i | X_{pa(i)}) = \prod_{(i,j) \in E} f_{ij}(X_i, X_j)$.

16. Polytrees (DAG) $\rightarrow$ Factor model.

Single rooted tree (DAG) $\rightarrow$ pairwise factor

pairwise
undirected tree $\rightarrow$ through pairwise.
marginals $P(X_i), P(X_i, X_j)$ from BP

17. Undirected tree to rooted directed trees.

- choose an arbitrary node X_r to be root and point all arrows away from it.

- compute $P(X_i)$ and $P(X_i, X_j)$.

$$P(X) = P(X_r) \prod_{i \neq r} P(X_i | X_{\text{par}(i)}) = P(X_r) \prod_{i \neq r} \frac{P(X_i, X_{\text{par}(i)})}{P(X_{\text{par}(i)})} = \frac{\prod_{\text{edges } (i,j)} P(X_i, X_j)}{\prod_{\text{nodes } i} P(X_i)^{\deg(i)-1}}$$

$\deg(i) = \# \text{ being parent} + \# \text{ being child}$. Only cancel parent case. For not $\deg(i) = \# \text{ being parent}$, leave one as prior. $\deg(i) = \text{number of neighbours of } i$.

18. Finding marginals in undirected trees

undirected tree $\Rightarrow$ pairwise factored $P(X) = \frac{1}{Z} \prod_{(i,j) \in E} f_{ij}(X_i, X_j)$.

$$P(X_i) = \sum_{x_1, x_2, \dots} P(X) \propto \sum_{x_1, x_2, \dots} \prod_{(i,j) \in E} f_{ij}(X_i, X_j) = \sum_{x_1, x_2, \dots} \prod_{j \in \text{necc}(i)} f_{ij}(X_i, X_j) \prod_{(i',j') \in E_{i' \rightarrow i}} f_{i'j'}(X_{i'}, X_{j'})$$

marginalize others

contain X_i

connect to X_i 's neighbour.

marginalize all but X_i

$$= \prod_{j \in \text{necc}(i)} \left(\sum_{x_j} f_{ij}(X_i, x_j) \prod_{(i',j') \in E_{i' \rightarrow i}} f_{i'j'}(X_{i'}, X_{j'}) \right)$$

$$= \prod_{j \in \text{necc}(i)} M_{j \rightarrow i}(X_i)$$

$$M_{j \rightarrow i}(X_i) = \sum_{x_j} f_{ij}(X_i, x_j) \prod_{(i',j') \in E_{i' \rightarrow i}} f_{i'j'}(X_{i'}, X_{j'}) = \sum_{x_j} f_{ij}(X_i, x_j) \left(\sum_{x_{j'}} \prod_{(i',j') \in E_{i' \rightarrow i}} f_{i'j'}(X_{i'}, X_{j'}) \right)$$

marginalize all but X_j

$$= \sum_{x_j} f_{ij}(X_i, x_j) \prod_{x_k \in \text{necc}(j) \setminus X_i} M_{k \rightarrow j}(X_j)$$

$$P(X_i, X_j) = \sum_{x_1, x_2, \dots} P(X) \propto \sum_{x_1, x_2, \dots} \prod_{(i,j) \in E} f_{ij}(X_i, X_j) = \prod_{x_k \in \text{necc}(i) \setminus X_j} M_{k \rightarrow i}(X_i) \prod_{x_k \in \text{necc}(j) \setminus X_i} M_{k \rightarrow j}(X_j) f_{ij}(X_i, X_j)$$

19. Message from observed leaf nodes are conditioned rather than marginalized:

To compute $P(X_i)$: $M_{a \rightarrow i}(X_i) = \sum_{x_a} f_{ai}(X_a, X_i) \propto \sum_{x_a} P(X_a | X_i) P(X_i)$.

To compute $P(X_i | X_a = a)$: $M_{a \rightarrow i}(X_i) = f_{ai}(X_a = a, X_i) = \sum_{x_a} \delta_a(x_a) f_{ai}(X_a, X_i) \propto P(X_a = a | X_i) P(X_i)$

20. Observed internal node partition the graph, messages propagate independently.

$$M_{b \rightarrow j} = f_{bj}(X_b = b, X_j) \quad M_{b \rightarrow k} = f_{bk}(X_b = b, X_k)$$

21. Messages $M_{i \rightarrow j}$ are proportional to the likelihood based on any observed variables ($\odot$) within the messages subtree $T_{i \rightarrow j}$, possibly scaled by a prior factor: $M_{i \rightarrow j}(X_j) \propto P(\mathcal{K}_{T_{i \rightarrow j}} \cap \odot | X_j) P(X_j)$.

22. BP for latent chain models: $\alpha_t(i) \Leftrightarrow M_{S_{t-1} \rightarrow S_t}(S_t = i) \quad M_{X_t \rightarrow S_t}(S_t = i) = M_{S_{t-1} \rightarrow S_t}(S_t = i) \sum_{x_t} P(x_t | S_t)$

$$\beta_t(i) \Leftrightarrow M_{S_{t+1} \rightarrow S_t}(S_t = i) \propto P(X_{t+1} | S_t)$$

$$\alpha_t(i) \beta_t(i) = \prod_{j \in \text{necc}(i)} M_{j \rightarrow S_t}(S_t = i) \propto P(S_t = i | \odot) \quad \odot = \{X_1, \dots, X_T\}$$

23. BP for polytrees: Associated undirected tree contains loops created by non-binary cliques. We can treat factors as message-passing nodes. $P(X) = \prod_i f_i(X_i) \prod_{a \in \mathcal{F}} f_a(X_{ca}) = \prod_{a \in \mathcal{F}} f_a(X_{ca})$

For factor $f_a(X_i, X_j, X_k, \dots)$

Incoming (node $\rightarrow$ factor) $M_{i \rightarrow a}(X_i) = \prod_{b \in \text{necc}(i) \cap a} M_{b \rightarrow i}(X_i)$

Outgoing (factor $\rightarrow$ node) $M_{a \rightarrow i}(X_i) = \sum_{x_a \setminus X_i} f_a(X_i, x_a \setminus X_i) \prod_{x_k \in \text{necc}(a) \setminus i} M_{k \rightarrow a}(X_k) \quad P(X_i) = \prod_{a \in \text{necc}(i)} M_{a \rightarrow i}(X_i)$

24. Transform from G to an easy-to-handle G' will only be valid if $P(X)$ can also be represented by G'
 Only remove conditional independence, never add them.

25. Junction tree algorithm:

① DAG $\rightarrow$ factor graph. $P(X) = \prod_i P(X_i | X_{pa(i)}) = \prod_i f_i(X_{C_i})$. $C_i = i \cup pa(i)$ root $P(X_r)$ absorbed into adjacent factor.
 observation in a factor graph $P(X|\theta) \propto P(X, \theta) = P(X) \mathbb{1}\{\theta\}$

So we can add singleton factor with δ -function: $P(X, \theta) = [\prod_a f_a] f_\theta(\theta) f_c(c)$ $\theta = \{D, C\}$
 $f_\theta(D) = \delta_d(D)$ $f_c(C) = \delta_c(C)$

② factor graph $\rightarrow$ undirected graph (moralization).

replace each factor by an undirected clique.

Construct potentials on each maximal clique by multiplying together factor potentials, ensure each factor potential only appears once.

③ Triangulation: add edges to the graph so every loop of size ≥ 4 has at least one chord

Variable elimination: A graph including all edges that would be induced by elimination is chordal (aka triangulated). Eliminate one node and connect its neighbours

Minimum deficiency search: choose variable that induces the fewest new edges. ✓

Maximum cardinality search: choose variable with most previously visited neighbours

Chord must be direct connections.

④ Chordal graph to junction tree.

Node represents a clique, edges are intersections of cliques, called separators.

Running intersection property: if two cliques contain variable X , all cliques and separators on the path between the two cliques contain X . This is guaranteed if graph is chordal.

26. Message passing on junction trees (node $i \leftrightarrow$ maximal clique C_i , edge $a \leftrightarrow$ separators $S_a = C_i \cap C_j$).



This can be viewed as (shaper-shenoy propagation)

$$C_i \text{ --- } S_{ij} \text{ --- } C_j \quad S_{ij} = C_i \cap C_j \quad M_{i \rightarrow j}(X_{S_{ij}}) = \sum_{X_{C_i} \setminus S_{ij}} f(C_i) \prod_{k \in \text{ne}(i) \setminus j} M_{k \rightarrow i}(X_{S_{ki}}).$$

$$P(X_{C_i}) \propto f_i(X_{C_i}) \prod_{k \in \text{ne}(i)} M_{k \rightarrow i}(X_{S_{ki}}). \quad P(X_{S_{ij}}) \propto M_{i \rightarrow j}(X_{S_{ij}}) M_{j \rightarrow i}(X_{S_{ij}}).$$

27. Local consistency $\rightarrow$ global consistency

If $q_i(X_{C_i})$, $r_{ij}(X_{S_{ij}})$ are distributions such that $\sum_{X_{C_i} \setminus S_{ij}} q_i(X_{C_i}) = r_{ij}(X_{S_{ij}})$.

(we also have $\sum_{X_{S_{ij}} \setminus C_j} r_{ij}(X_{S_{ij}}) = q_j(X_{C_j})$), then $P(X) = \frac{\prod_{\text{clique } i} q_i(X_{C_i})}{\prod_{\text{separator } ij} r_{ij}(X_{S_{ij}})}$ is also a distribution

with $q_i(X_{C_i}) = \sum_{X_{C_i}} P(X)$ and $r_{ij}(X_{S_{ij}}) = \sum_{X_{S_{ij}}} P(X)$.

We need the division of separator potential as $C_i \cap C_j$ is counted twice.

28. Hugin propagation:

- Initialise $q_i(X_{C_i}) \propto f_i(X_{C_i})$ $r_{ij}(X_{S_{ij}}) \propto 1$, then $P(X) \propto \frac{\prod_i q_i(X_{C_i})}{\prod_{ij} r_{ij}(X_{S_{ij}})}$

- update for $i \rightarrow j$: $r_{ij}^{\text{new}}(X_{S_{ij}}) = \sum_{X_{C_i} \setminus S_{ij}} q_i(X_{C_i})$, $q_j^{\text{new}}(X_{C_j}) = q_j(X_{C_j}) \frac{r_{ij}^{\text{new}}(X_{S_{ij}})}{r_{ij}(X_{S_{ij}})}$

$P(X)$ is always unchanged. local consistency.

29. Running and memory costs of message passing are both $O(\sum_i |X_{C_i}| |X_{C_i}|)$ number of combination of states in clique i .

30. Learning in graphical models: $L = P(X_{\text{obs}} | \theta) = \sum_{X_{\text{unobs}}} P(X_{\text{obs}}, X_{\text{unobs}} | \theta)$

$$F(q, \theta) = \langle \log P(X_{\text{obs}}, X_{\text{unobs}} | \theta) - \log q(X_{\text{unobs}}) \rangle_{q(X_{\text{unobs}})} = \langle \sum_i \log f_i(X_{C_i} | \theta) - \log Z(\theta) \rangle_{q(X_{\text{unobs}})} + H(q)$$

$$= \sum_i \langle \log f_i(X_{C_i} | \theta) \rangle_{q(X_{C_i} \setminus X_{\text{obs}})} - \log Z(\theta) + H(q).$$

31. Bayesian evidence: $P(D|m) = \int_{\Theta_m} P(D|\theta_m, m) P(\theta_m|m) d\theta_m$ m is model

In conjugate prior and exponential family case, $P(D|m) = \frac{\text{posterior volume}}{\text{prior volume}}$.

32. Laplace approximation: $P(D, \theta_m|m) \propto P(\theta_m|D, m)$

Approximate $\log P(D, \theta_m|m)$ to second-order around θ^* (mode from posterior).

$$\log P(D, \theta_m|m) \approx \log P(D, \theta_m^*|m) + \nabla_{\theta_m} \log P(D, \theta_m^*|m) (\theta_m - \theta_m^*) + \frac{1}{2} (\theta_m - \theta_m^*)^T \nabla_{\theta_m}^2 \log P(D, \theta_m^*|m) (\theta_m - \theta_m^*)$$

$\nabla \log P(D, \theta_m^*|m) = 0$ and let $A := -\nabla^2 \log P(D, \theta_m^*|m)$ (negative Hessian of $\log P(D, \theta|m)$ evaluated at θ_m^*)

$$\text{So } P(D|m) = \int e^{\log P(D, \theta_m|m)} d\theta_m \approx \int P(D, \theta_m^*|m) e^{-\frac{1}{2} (\theta_m - \theta_m^*)^T A (\theta_m - \theta_m^*)} d\theta_m$$

$$= P(D|\theta_m^*, m) P(\theta_m^*|m) (2\pi)^{\frac{d}{2}} |A|^{-\frac{1}{2}}$$

approximating posterior with Gaussian.

$$33. \log P(D|m) \approx \log P(\theta_m^*|m) + \log P(D|\theta_m^*, m) + \frac{d}{2} \log 2\pi - \frac{1}{2} \log |A|$$

$N \rightarrow \infty$, $A \rightarrow NA_0 + \text{const}$ A_0 is a fixed positive definite matrix $\propto -\nabla^2 \log P(X|\theta^*, m) >$

$$\log |A| \rightarrow \log |NA_0| = \log (N^d |A_0|) = d \log N + \log |A_0|$$

So leave only terms growing with N :

$$\log P(D|m) \approx \underbrace{\log P(D|\theta_m^*, m)}_{\text{log likelihood}} - \underbrace{\frac{d}{2} \log N}_{\text{penalty}} \quad (\text{BIC} = d \log N - 2 \log P(D|\theta_m^*, m))$$

$$(\text{AIC} = 2d - 2 \log P(D|\theta_m^*, m))$$

we can use ML estimate of θ instead of MAP.

$$34. W \sim \mathcal{N}(0, C), y_i \sim \mathcal{N}(w^T x_i, \sigma^2) \text{ posterior on } w: \mathcal{N}(\sum_w \frac{X Y^T}{\sigma^2}, (\underbrace{\frac{X X^T}{\sigma^2} + C^{-1}}_{\text{data precision} + \text{prior precision}})^{-1})$$

If we take $C^{-1} = \text{diag}(\alpha)$ (i.e. $w_i \sim \mathcal{N}(0, \alpha_i^{-1})$),

$\mathcal{E}(C, \sigma^2) = \int P(Y|X, w, C, \sigma^2) P(w|C) dw$ to get evidence. and take derivative

$$\alpha_i^{\text{new}} = \frac{1 - \alpha_i [\sum_w]_{ii}}{\bar{w}_i^2} \quad (\sigma^2)^{\text{new}} = \frac{(Y - \bar{w}^T X)(Y - \bar{w}^T X)^T}{N - \sum_i (1 - [\sum_w]_{ii} \alpha_i)}$$

$\alpha_i \rightarrow \infty \Rightarrow w_i = 0$ irrelevant input x_i
 α_i finite $\Rightarrow w_i = \arg \max P(w_i | X, Y, \alpha_i)$ relevant input x_i

Automatic Relevance Determination (ARD), yields sparse solutions

35. Prediction averaging: $P(x|D, m) = \int P(x|\theta, m) P(\theta|D, m) d\theta$ (unsupervised learning)

$P(y|x, D, m) = \int P(y|x, \theta, m) P(\theta|D, m) d\theta$ (supervised learning)

The integral naturally favours predictions associated with large volumes in posterior, so incorporates Occam-like factor.

36. Bayesian nonparametrics: marginalized model has no parameters.

37. If $w \sim \mathcal{N}(0, \tau^2 I)$ $y_i \sim \mathcal{N}(w^T x_i, \sigma^2)$, integrate w to get $y|x, D \sim \mathcal{N}(\bar{w}^T x, x^T \sum_w x + \sigma^2)$ use posterior w

$$\mathbb{E}[y|x, D] = \mathbb{E}[\mathbb{E}[y|x, w]] = \bar{w}^T x \quad \text{var}(y|x, D) = \mathbb{E}[\text{var}(y|x, w)] + \text{var}(\mathbb{E}[y|x, w]) = x^T \sum_w x + \sigma^2$$

If integrate w with prior one, we get $y \sim \mathcal{N}(0, \tau^2 X^T X + \sigma^2 I)$

$$38. \begin{bmatrix} Y^T \\ y \end{bmatrix} | X, x \sim \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tau^2 X^T X + \sigma^2 I & \tau^2 X^T x \\ \tau^2 x^T X & \tau^2 x^T x + \sigma^2 \end{bmatrix} \right) = \mathcal{N} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \tilde{K}_{xx} & \tilde{K}_{x} \\ \tilde{K}_{x} & \tilde{K}_{xx} \end{bmatrix} \right)$$

$$\text{so } y | Y, X, x \sim \mathcal{N}(\tilde{K}_{xx} \tilde{K}_{xx}^{-1} Y^T, \tilde{K}_{xx} - \tilde{K}_{xx} \tilde{K}_{xx}^{-1} \tilde{K}_{xx}) \quad \text{explain away from regression.}$$

$$\sim \mathcal{N}(x^T \underbrace{\frac{1}{\sigma^2} \sum X Y^T}_{\tilde{\Sigma}}, x^T \underbrace{\sum x + \sigma^2}_{\tilde{\Sigma}}) \quad \tilde{\Sigma} = (\frac{1}{\sigma^2} X X^T + \frac{1}{\tau^2} I)^{-1}$$

Same as we integrate posterior w .

39. Posterior-integration $\equiv$ Joint Gaussian with kernel and do conditioning.
 (parameter integrated out) (parameter free).

40. covariance kernel $\tilde{K}$: $Y|X, \tilde{K} \sim \mathcal{N}(0, \tilde{K}_{xx})$

evidence optimisation: gradient ascent in $\log P(Y|X, \tilde{K})$

41. Gaussian Process: a stochastic process on $\mathbb{R}$ indexed by $x \in \mathcal{X}$, any subset of which have Gaussian distributions. GP is defined by a kernel and mean function.

A random function $f: \mathcal{X} \rightarrow \mathbb{R}$, we write $f(\cdot) \sim \text{GP}(m(\cdot), K(\cdot, \cdot))$.

$f = [f(x_1), f(x_2), \dots, f(x_N)]^T$ with points $[x_1, \dots, x_N]$ $f|X, K \sim \mathcal{N}(m, K_{xx})$.

42. Brownian motion a Gaussian process with kernel function $k(s, t) = \min(s, t)$.

43. Regression with Gaussian Process.

- GP prior: $f(\cdot) \sim \text{GP}(0, K(\cdot, \cdot))$, $f(x) \sim \mathcal{N}(0, K_{xx})$. $\begin{bmatrix} f(x) \\ f(x) \end{bmatrix} \sim \mathcal{N}(0, \begin{bmatrix} K_{xx} & K_{xx} \\ K_{xx} & K_{xx} \end{bmatrix})$

- $y_i | x_i, f \sim \mathcal{N}(f(x_i), \sigma^2)$ so $Y \sim \mathcal{N}(0, \tilde{K}_{xx})$ with $\tilde{K}_{xx} = K_{xx} + \sigma^2 I$. $\begin{bmatrix} Y \\ f(x) \end{bmatrix} \sim \mathcal{N}(0, \begin{bmatrix} \tilde{K}_{xx} & K_{xx} \\ K_{xx} & K_{xx} \end{bmatrix})$

Posterior on latent f is also a GP

$f(\cdot) | X, Y \sim \text{GP}(K_{\cdot x} (K_{xx} + \sigma^2 I)^{-1} Y^T, K(\cdot, \cdot) - K_{\cdot x} (K_{xx} + \sigma^2 I)^{-1} K_{x \cdot})$

Prediction: posterior on f , plus observation noise

$y | X, Y, x \sim \mathcal{N}(\mathbb{E}[f(x) | Y, X], \text{Var}[f(x) | Y, X] + \sigma^2) = \mathcal{N}(K_{xx} \tilde{K}_{xx}^{-1} Y, K_{xx} - K_{xx} \tilde{K}_{xx}^{-1} K_{xx} + \sigma^2)$

44. A GP with a translation-invariant covariance kernel is stationary: if $f(\cdot) \sim \text{GP}(0, K)$, then so is $f(\cdot - x) \sim \text{GP}(0, K)$ for each x .

45. A covariance kernel is radial or radially symmetric if $K(x, x') = h(\|x - x'\|)$.

A GP with a radial covariance kernel is stationary with respect to translations, rotations and reflections of the input space.

46. Independent Components Analysis (ICA): $X_d = \sum_{k=1}^K \Lambda_{dk} Z_k + E_d$ $X = \Lambda Z + E$
 $Z_k \stackrel{\text{iid}}{\sim} P_Z$ non-Gaussian
 well-posed even with $K \gg D$
 Tractable for 0 noise.

47. Square, noiseless ICA (also called infomax ICA) $X = \Lambda Z \Rightarrow Z = W X$ with $W = \Lambda^{-1}$

Z is independent component. W is the unmixing matrix.

If we have $x = F(z)$ bijectively, we have $f_x(x) = f_z(F^{-1}(x)) \left| \frac{dF^{-1}(x)}{dx} \right|$

So $P(X|W) = \prod_k P_Z(\underbrace{[WX]_k}_{z_k}) |W|$ Z s are independent, so $P(Z) = \prod_{k=1}^K P_{Z_k}(Z_k)$.

$\log P(X) = \log |W| + \sum_i \log P_Z(W_i X) \Rightarrow \Delta W \propto \nabla_W \log P(X) = (W^{-1})^T + g(Z) X^T$ $g(Z) = \frac{\partial \log P(Z)}{\partial Z}$

Better approach: "natural" or covariant gradient $\Delta W \propto \nabla_W \log P(X) \cdot (W^T W) = W + g(Z) Z^T W \approx \langle -\nabla \log P \rangle^T$

Cannot use EM because no randomness, the posterior is a dirac delta function.

$\mathbb{E}[S|X, A]$ is trivially $s = A^{-1}x$ and A is recovered from s in M steps. A circular

48. Mutual Information: $I(X; Y) = \text{KL}[P_{XY} || P_X P_Y] = H(X) - H(X|Y) = H(Y) - H(Y|X)$.

49. Conditional Entropy: $H(Y|X) = - \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} P(x, y) \log \frac{P(x, y)}{P(x)}$

50. A feedforward model: $z_i = W_i X$; $z_i = f_i(z_i)$ with $f_i(-\infty) = 0$ $f_i(+\infty) = 1$ (squashing function)

Infomax: $\arg \max_W I(X; Z) = \arg \max_W (H(Z) - H(Z|X)) = \arg \max_W H(Z)$ because $f(W_i X)$ is a deterministic function given x fixed.
 maximize information carried by z about x .
 so $H(Z|X) = 0$.

Maximize entropy of z , making it uniform as possible as on $[0, 1]$ forces rows of W to become statistically independent sources.

Subadditivity of entropy: $H(Z) \leq \sum_i H(z_i)$ with equality $\Leftrightarrow$ components are statistically independent.
 uniform on $[0, 1] \Rightarrow$ max entropy for z_i statistically independent fill the gap for global entropy
 $H(Z) = \sum_i H(z_i)$.

$H(Z) = \sum_i H(z_i) - I(z_1, \dots, z_D)$. Infomax ICA $\Leftrightarrow$ square noiseless causal ICA $z_i = f_i(z_i) = \text{cdf}_i(z_i)$ $W = \Lambda^{-1}$

51. Kurtosis = $\frac{E[(X-\mu)^4]}{E[(X-\mu)^2]^2} - 3$. Linear mixtures of independent non-Gaussian sources tend to be more Gaussian $\Rightarrow k \rightarrow 0$ (Central limit theorem).

52. Nonlinear state-space model (NLSSM):

Extended Kalman Filter (EKF): linearize nonlinear functions about current estimate $\hat{z}_t^t$.

$z_{t+1} = f(z_t, u_t) + w_t$, $x_t = g(z_t, u_t) + v_t$. w_t, v_t usually still Gaussian.

$$z_{t+1} \approx \underbrace{f(\hat{z}_t^t, u_t)}_{\tilde{B}_t u_t} + \underbrace{\frac{\partial f}{\partial z_t} \bigg|_{\hat{z}_t^t}}_{\tilde{A}_t} (z_t - \hat{z}_t^t) + w_t \quad x_t \approx \underbrace{g(\hat{z}_t^t, u_t)}_{\tilde{D}_t u_t} + \underbrace{\frac{\partial g}{\partial z_t} \bigg|_{\hat{z}_t^t}}_{\tilde{C}_t} (z_t - \hat{z}_t^t) + v_t$$

$$z_{t+1} \approx \tilde{B}_t u_t + \tilde{A}_t (z_t - \hat{z}_t^t) + w_t \quad x_t \approx \tilde{D}_t u_t + \tilde{C}_t (z_t - \hat{z}_t^t) + v_t$$

u_t is just an argument with a fixed realized value. treat it as given/observed!

So we can write $\tilde{z}_{t+1} \approx \tilde{A}_t \tilde{z}_t^t + \tilde{w}_t$, $x_t \approx \tilde{C}_t \tilde{z}_t^t + \tilde{v}_t$ $\tilde{z}_t^t = z_t - \hat{z}_t^t$, $\tilde{z}_{t+1}^t = z_{t+1} - \hat{z}_{t+1}^t$

when we make prediction, we use $f(\hat{z}_t^t, u_t) = \hat{z}_{t+1}^t$ $\hat{x}_{t+1}^t = g(\hat{z}_{t+1}^t, u_{t+1})$.

53. Boltzmann Machines: undirected graphical model over binary variables $S_i \in \{0, 1\}$. Some S_i are hidden, some are observed.

$$P(S|w, b) = \frac{1}{Z} \exp \left\{ \sum_{i,j} w_{ij} S_i S_j - \sum_i b_i S_i \right\} \quad w_{ij} = w_{ji}$$

Its energy function $H(S) = -\sum_{i,j} w_{ij} S_i S_j + \sum_i b_i S_i$ such that $P(S) = \frac{1}{Z} e^{-H(S)}$.

Inference requires $\langle S^H \rangle_{P(S^H|S^V, w, b)}$ and $\langle S^H S^H \rangle_{P(S^H|S^V, w, b)}$.

Learning: $\log P(S^H S^V | w, b) = \sum_{i,j} w_{ij} S_i S_j - \sum_i b_i S_i - \log Z$. $Z = \sum_S \exp(\sum_{i,j} w_{ij} S_i S_j - \sum_i b_i S_i)$.

In M-step we have $\frac{\partial F}{\partial \theta} = \frac{\partial}{\partial w_{ij}} \langle \log P(S^H S^V | w, b) \rangle_{P(S^H|S^V, w, b)} = \langle S_i S_j \rangle_c - \frac{\partial}{\partial w_{ij}} \log Z$

$\langle \rangle_c$ means take expectation on posterior $P(S^H|S^V, w, b)$, use $S^V = V_{obs}$. Clamped expectation $= \langle S_i S_j \rangle_c - \sum_S P(S|w, b) S_i S_j$

$\langle \rangle_u$ means full model $P(S|w, b)$. Unclamped expectation $= \langle S_i S_j \rangle_c - \langle S_i S_j \rangle_u$

54. Restricted Boltzmann machines: $w_{ij} = 0$ for any two visible or invisible nodes. (bipartite graph)

$$P(S^V | S^H) = \frac{1}{Z} \exp(\sum_{i \in V} \sum_{j \in H} w_{ij} S_i S_j - \sum_{i \in V} b_i S_i - \sum_{j \in H} b_j S_j)$$

$$= \frac{1}{Z} \prod_{i \in V} \exp(S_i (\sum_{j \in H} w_{ij} S_j - b_i)) = \prod_{i \in V} \text{Bernoulli}(\sigma(\sum_{j \in H} w_{ij} S_j - b_i))$$

$$P(S^H | S^V) = \prod_{j \in H} \text{Bernoulli}(\sigma(\sum_{i \in V} w_{ij} S_i - b_j))$$

Unclamped samples can be generated efficiently by block Gibbs sampling.

55. Latent Dirichlet Allocation (LDA model)

W possible words D documents K topics

LDA model specifies distribution of i -th word in d -th document, $x_{id} \in \{1, \dots, W\}$

Topic distribution for d -th document: $\theta_d | \alpha \sim \text{Dirichlet}(\alpha, \dots, \alpha)$ length K

Word distribution for k -th topic: $\phi_k | \beta \sim \text{Dirichlet}(\beta, \dots, \beta)$ length W .

Topic for i -th word in d -th document: $z_{id} | \theta_d \sim \text{Discrete}(\theta_d)$ length K

Identity of i -th word in d -th document: $x_{id} | z_{id}, \phi_{z_{id}} \sim \text{Discrete}(\phi_{z_{id}})$ Given a topic, length W .

P_{dw} is the probability of word w appearing in document d .

$$P_{dw} = \sum_{k=1}^K \theta_{dk} \phi_{kw} = \underbrace{D \left[\begin{matrix} K \\ \alpha \end{matrix} \right]}_{\text{Dirichlet}} \underbrace{K \left[\begin{matrix} W \\ \beta \end{matrix} \right]}_{\text{Dirichlet}}$$

56. In variational approximation, if $P(Z|X, \theta^M) \in \mathcal{Q}$, $\mathcal{Q}$ is the class of approximation, then θ^M is a fixed point of the variational algorithm.

$$57. F(q, \theta) = \langle \log P(X, Z|\theta) \rangle_{q(z)} + H[q] = \langle \log P(X|\theta) \rangle_{q(z)} - KL[q \| P(Z|X, \theta)] \\ = \log P(X|\theta) - KL[q \| P(Z|X, \theta)]$$

$$\text{Estep: } q^{(k)}(z) = \arg\max_{q(z) \in \mathcal{Q}} F(q(z), \theta^{(k-1)}) \Leftrightarrow q^{(k)}(z) := \arg\min_{q(z) \in \mathcal{Q}} KL[q(z) \| P(Z|X, \theta^{(k-1)})]$$

58. Factored Variational E-step.

Most common one $\mathcal{Q} = \{q | q(z) = \prod_i q_i(z_i)\}$ disjoint sets Z_i

In i -th step of E: maximise $F(q, \theta)$ wrt $q_i(z_i)$ only, keep other q_j and parameters.

$$q_i^{(k)}(z_i) = \arg\max_{q_i(z_i)} F(q_i(z_i) \prod_{j \neq i} q_j(z_j), \theta^{(k-1)})$$

Iterate until converge in E.

$$F(\prod_j q_j(z_j), \theta^{(k-1)}) = \langle \log P(X, Z|\theta^{(k-1)}) \rangle_{\prod_j q_j(z_j)} + H[\prod_j q_j(z_j)] = \int q_i(z_i) \langle \log P(X, Z|\theta^{(k-1)}) \rangle_{\prod_{j \neq i} q_j(z_j)} dz_i \\ + H[q_i] + \sum_{j \neq i} H[q_j]$$

$$\frac{\partial}{\partial q_i} (F + \lambda (\int q_i - 1)) = \langle \log P(X, Z|\theta^{(k-1)}) \rangle_{\prod_{j \neq i} q_j(z_j)} - \log q_i - 1 + \lambda = 0$$

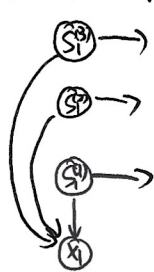
$$\Rightarrow q_i(z_i) \propto \exp \langle \log P(X, Z|\theta^{(k-1)}) \rangle_{\prod_{j \neq i} q_j(z_j)}$$

59. Mean-field approximations: $Z_i = z_i$ (q is factored to each variables).

60. In Boltzmann machine, $P(X, Z) = \frac{1}{Z} \exp(\sum_j w_{ij} s_i s_j + \sum_i b_i s_i)$.

$$\langle \log P(X, Z|\theta) \rangle_{\pi_{z_i}} = \sum_j w_{ij} \langle s_i \rangle_{z_i} \langle s_j \rangle_{z_j} + \sum_i b_i \langle s_i \rangle_{z_i} \quad (\text{for } s_i \text{ in } i\text{-th step, we have } s_i \sum_j w_{ij} \langle s_j \rangle_{z_j} \text{ mean field.})$$

61. Mean-field FHMM



$$q(S_{1:T}^{1:m}) = \prod_{t=1}^T q_t^m(S_t^m)$$

$$q_t^m(S_t^m) \propto \exp \langle \log P(S_{1:T}^{1:m}, X_{1:T}) \rangle_{\prod_{t'=1}^m q_{t'}^m} = \exp \langle \sum_{\mu} \sum_t \log P(S_t^\mu | S_{t-1}^\mu) + \sum_t \log P(X_t | S_t^\mu) \rangle_{\prod_{t'=1}^m q_{t'}^m}$$

$$\propto \exp \left[\langle \log P(S_t^m | S_{t-1}^m) \rangle_{\prod_{t'=1}^m q_{t'}^m} + \langle \log P(X_t | S_t^m) \rangle_{\prod_{t'=1}^m q_{t'}^m} + \langle \log P(S_{t+1}^m | S_t^m) \rangle_{\prod_{t'=1}^m q_{t'}^m} \right]$$

$$\alpha_t^m(i) \propto \exp(\sum_j \log \Phi_{ji}^m q_{t-1}^m(j) + \langle \log A_i(X_t) \rangle_{q_t^m}) \quad \beta_t^m(i) \propto \exp(\sum_j \log \Phi_{ij}^m q_{t+1}^m(j))$$

We know $q_t^m(S_{t-1}^m) \propto \alpha_t^m(i) \beta_t^m(i)$

62. Structured FHMM

$$q(S_{1:T}^{1:m}) = \prod_{t=1}^T q_t^m(S_t^m)$$

$$q_t^m(S_{1:T}^m) \propto \exp \langle \log P(S_{1:T}^{1:m}, X_{1:T}) \rangle_{\prod_{t'=1}^m q_{t'}^m(S_{1:T}^{t'})} \propto \exp \left[\sum_t \log P(S_t^m | S_{t-1}^m) + \sum_t \langle \log P(X_t | S_t^m) \rangle_{\prod_{t'=1}^m q_{t'}^m} \right]$$

$$= \prod_t P(S_t^m | S_{t-1}^m) \prod_t \exp(\langle \log P(X_t | S_t^m) \rangle_{\prod_{t'=1}^m q_{t'}^m})$$

63. For DAG: $P(X, Z) = \prod_k P(V_k | \text{pa}(V_k))$, let $q(z) = \prod_i q_i(z_i)$ for disjoint sets $\{Z_i\}$.

$$\log q_i^*(Z_i) = \langle \sum_k \log P(V_k | \text{pa}(V_k)) \rangle_{q_i(Z_i)} + \text{const} = \sum_{j \in \text{parent}(i)} \langle \log P(Z_j | \text{pa}(Z_j)) \rangle_{q_j(Z_j)} + \sum_{j \in \text{children}(i)} \langle \log P(V_j | \text{pa}(V_j)) \rangle_{q_j(Z_j)} + \text{const}$$

Each node receives message from Markov boundary.

64. Variational Bayes:

$$\log P(X|\mu) = \log \iint P(X, Z|\theta, \mu) P(\theta|\mu) dZ d\theta \geq \iint Q(Z, \theta) \log \frac{P(X, Z|\theta, \mu)}{Q(Z, \theta)} dZ d\theta$$

$$\geq \iint \log \frac{P(X, Z|\theta, \mu)}{Q_Z(Z) Q_\theta(\theta)} Q_Z(Z) Q_\theta(\theta) dZ d\theta$$

(Evidence Lower bound, ELBO).

$$F(Q_Z, Q_\theta) = \iint \log \frac{P(X, Z|\theta, \mu)}{Q_Z(Z) Q_\theta(\theta)} Q_Z(Z) Q_\theta(\theta) dZ d\theta$$

$$\text{E-like step: } Q_Z^*(Z) \propto \exp \langle \log P(Z, X|\theta) \rangle_{Q_\theta(\theta)}$$

$$\text{M-like step: } Q_\theta^*(\theta) \propto \exp \langle \log P(Z, X|\theta) \rangle_{Q_Z(Z)} \cdot P(\theta)$$

$$\log P(X) - F(Q_Z, Q_\theta) = KL(Q_Z(Z) Q_\theta(\theta) \| P(Z, \theta|X))$$

65. Conjugate-exponential VB

$Q_\theta(\theta)$ is also conjugate: $Q_\theta(\theta) \propto P(\theta) \exp \langle \log P(Z, X | \theta) \rangle_{Q_\theta}$

$$= h(\nu, \tau) g(\theta)^\nu \exp(\phi(\theta)^\top \tau) \cdot g(\theta)^\tau \exp(\langle \log f(z, x) \rangle_{Q_\theta} + \phi(\theta)^\top \tau \sum_i T(z_i, x_i) \rangle_{Q_\theta}) \\ \propto h(\tilde{\nu}, \tilde{\tau}) g(\theta)^\tilde{\nu} \exp(\phi(\theta)^\top \tilde{\tau}) \quad \tilde{\nu} = \nu + n \quad \tilde{\tau} = \tau + \sum_i \langle T(z_i, x_i) \rangle_{Q_\theta}$$

only need to track $\tilde{\nu}, \tilde{\tau}$

$$Q_Z(Z) = \prod_i Q_{z_i}(z_i) \cdot Q_{\tilde{\tau}}(\tilde{\tau}) \propto \exp \langle \log P(z_i, x_i | \theta) \rangle_{Q_\theta} \propto f(x_i, z_i) \exp(\langle \phi(\theta) \rangle_{Q_\theta}^\top T(z_i, x_i)) \\ = P(z_i | x_i, \bar{\phi}(\theta))$$

change natural parameter from $\phi(\theta)$ to $\langle \phi(\theta) \rangle_{Q_\theta}$

66. EM for MAP v.s. VB EM

EM for MAP estimation

Goal: maximize $P(\theta | \mathcal{X}, n)$ wrt θ

E step: compute $Q_Z(Z) \leftarrow P(Z | \mathcal{X}, \theta)$

M step: $\theta \leftarrow \arg \max_\theta \int \log P(Z, \mathcal{X}, \theta) Q_Z(Z) dZ$

Variational Bayesian EM

Goal: Maximize bound on $P(\mathcal{X} | n)$ wrt Q_θ

VB-E step: compute $Q_Z(Z) \leftarrow P(Z | \mathcal{X}, \bar{\phi})$

VB-M step: $Q_\theta(\theta) \leftarrow \exp \int \log P(Z, \mathcal{X}, \theta) Q_Z(Z) dZ$

- Reduce to EM if $Q_\theta(\theta) = \delta(\theta - \theta^*)$, in which case $\langle \phi(\theta) \rangle_{Q_\theta(\theta^*)} = \phi(\theta^*)$.

$$67. F(Q_Z, Q_\theta, \eta) = \iint Q_Z(Z) Q_\theta(\theta) \log \frac{P(\mathcal{X}, Z, \theta | \eta)}{Q_Z(Z) Q_\theta(\theta)} dZ d\theta \leq P(\mathcal{X} | \eta)$$

hyper-M step maximises current bound wrt η . $\eta \leftarrow \arg \max_\eta \iint Q_Z(Z) Q_\theta(\theta) \log P(\mathcal{X}, Z, \theta | \eta) dZ d\theta$.

68. ARD with variational Bayesian methods for latent dimensionality.

eg. $x \sim N(\Lambda z, \Psi)$ $z \sim N(0, I)$, column-wise prior $\Lambda_i \sim N(0, \alpha_i^{-1} I)$.

$$F(Q_Z, Q_\Lambda, \Psi, \alpha) = \langle \log P(\mathcal{X}, Z | \Lambda, \Psi) + \log P(\Lambda | \alpha) + \log P(\Psi) \rangle_{Q_Z Q_\Lambda} + \text{const.}$$

$\alpha \leftarrow \arg \max_\alpha \langle \log P(\Lambda | \alpha) \rangle_{Q_\Lambda}$ optimize α to cause some α_i diverge as ARD.

69. Variational Sparse GP approximations: Computing $N \times N$ matrix inversion takes $O(N^3)$ time.

Select a smaller set of possibly fictitious measurements U at input Z such that:

$$P(y' | Z, U, x') \approx P(y' | x, y, x')$$

F : latent GP values for X ; U : latent GP values for Z . GP prior: $\begin{bmatrix} F \\ U \end{bmatrix} \sim N(0, \begin{bmatrix} K_{XX} & K_{XZ} \\ K_{ZX} & K_{ZZ} \end{bmatrix})$

$$Y | F \sim N(F, \sigma^2 I)$$

$$\text{marginal likelihood } P(Y | X) = \int P(Y | F) P(F | X) dF = \iint \underbrace{P(Y | F)}_{\text{likelihood}} \underbrace{P(F | U, X, Z)}_{\text{GP conditional}} \underbrace{P(U | Z)}_{\text{GP prior at } Z} dF dU.$$

both U and F are latent, we introduce variational distribution $q(F, U)$

$$F(q(F, U), \theta) = \langle \log \frac{P(Y | F) P(F | U, X, Z) P(U | Z)}{q(F, U)} \rangle_{q(F, U)}. \quad \text{We choose } q(F, U) = \frac{P(F | U, X, Z) q(U)}{P(U)}$$

Fix $F | U$ without reference to Y , so information about Y will need to be "compressed" into $q(U)$.

$$\text{So } F(q(F, U), \theta, Z) = \langle \langle \log P(Y | F) \rangle_{P(F | U)} + \log P(U | Z) - \log q(U) \rangle_{q(U)}.$$

$$\langle \log P(Y | F) \rangle_{P(F | U)} = \langle -\frac{1}{2} \log |2\pi\sigma^2 I| - \frac{1}{2\sigma^2} \text{Tr}[(Y - F)(Y - F)^T] \rangle_{P(F | U)} \quad (\text{because we have } \langle FF^T \rangle_{P(F | U)} = \langle F \rangle_{P(F | U)} \langle F \rangle_{P(F | U)}^T + \Sigma_{F | U}) \\ = -\frac{1}{2} \log |2\pi\sigma^2 I| - \frac{1}{2\sigma^2} \text{Tr}[(Y - \langle F \rangle_{P(F | U)}) (Y - \langle F \rangle_{P(F | U)})^T] - \frac{1}{2\sigma^2} \text{Tr}[\Sigma_{F | U}] \\ = \log N(Y | K_{XZ} K_{ZZ}^{-1} U, \sigma^2 I) - \frac{1}{2\sigma^2} \text{Tr}[K_{XX} - K_{XZ} K_{ZZ}^{-1} K_{ZX}].$$

$$F(q(U), \theta, Z) = \langle \log \frac{N(Y | K_{XZ} K_{ZZ}^{-1} U, \sigma^2 I) P(U | Z)}{q(U)} \rangle_{q(U)} - \frac{1}{2\sigma^2} \text{Tr}[K_{XX} - K_{XZ} K_{ZZ}^{-1} K_{ZX}]$$

Linear Gaussian model: $U \sim N(0, K_{ZZ})$ (PPCA) $F | U \sim N(WU, \sigma^2 I)$

F is maximized at $q^*(U) = P(U | Y)$.

$$F(q^*(U), \theta, Z) = \log N(Y | 0, K_{XZ} K_{ZZ}^{-1} K_{ZX} + \sigma^2 I) - \frac{1}{2\sigma^2} \text{Tr}[K_{XX} - K_{XZ} K_{ZZ}^{-1} K_{ZX}]. \quad \text{optimize wrt } Z \text{ and } \theta.$$

$\log P(Y | U)$ computational cost reduces to M related

70. Sigma-point filter.

We have $\tilde{p}(z_{t+1} | x_{1:t+1})$ with mean $\hat{z}_{t+1}^{t+1}$ and variance matrix $\hat{V}_{t+1}^{t+1}$

Evaluate $f(\hat{z}_{t+1}^{t+1})$, $f(\hat{z}_{t+1}^{t+1} \pm \sqrt{\lambda} V)$ λ, V are $\hat{V}_{t+1}^{t+1} V = \lambda V$ for K dimensions.

Fit a Gaussian to these $2K+1$ σ -points to estimate $z_t = f(z_{t+1}) + \varepsilon$.

$$\mathcal{N}\left(\frac{1}{2K+1} \sum_{i=1}^{2K+1} f(\sigma_i), \frac{1}{2K+1} \sum_{i=1}^{2K+1} f(\sigma_i) f(\sigma_i)^T - \mu \mu^T + Q\right) \leftarrow \text{noise}$$

71. The other KL

$$\begin{aligned} \arg \min_{q_i} KL[p(z|x) \parallel \prod_i q_i(z_i|x)] &= \arg \min_{q_i} - \int p(z|x) \log \prod_i q_i(z_i|x) dz \\ &= \arg \min_{q_i} - \sum_i \int p(z|x) \log q_i(z_i|x) dz \\ &= \arg \min_{q_i} - \int p(z_i|x) \log q_i(z_i|x) dz \leftarrow \text{Cross entropy} = \\ &= H[p_i] + KL(p_i \parallel q_i) \end{aligned}$$

72. EP: $p(z|x) = \frac{1}{Z} \prod_i p(z_i | p_a(z_i)) \propto \prod_i f_i(z_i)$ z_i are not necessarily disjoint f_i are called sites
 q with same factorisation, but approximate sites $q(z) \propto \prod_i \tilde{f}_i(z_i)$.

Minimise $KL[p \parallel q] \Rightarrow \min_{\tilde{f}_i} KL[\tilde{f}_i(z_i) \prod_{j \neq i} \tilde{f}_j(z_j) \parallel \prod_{j \neq i} f_j(z_j) \prod_{j \neq i} \tilde{f}_j(z_j)]$ (other factors also tell message about z_i).

If z_i are truly disjoint, we don't need $\prod_{j \neq i} \tilde{f}_j(z_j)$. EP reduces to local KL problem.

$$\begin{aligned} 73. \tilde{f}_i^{\text{new}}(z) &\leftarrow \arg \min_{\tilde{f} \in \{\tilde{f}_j\}} KL[\tilde{f}_i(z) q_{T_i}(z) \parallel f(z_i) q_{T_i}(z)] \quad q_{T_i}(z) := \prod_{j \neq i} \tilde{f}_j(z_j). \\ &= \arg \min_{\tilde{f} \in \{\tilde{f}_j\}} KL[\tilde{f}_i(z_i) q_{T_i}(z_i) \parallel f(z_i) q_{T_i}(z_i)] \quad q_{T_i}(z_i) \text{ is cavity distribution.} \end{aligned}$$

74. EP algorithm: Input $f_1(z_1), \dots, f_N(z_N)$. Initialize $\tilde{f}_i(z_i) = \arg \min_{\tilde{f}} KL[\tilde{f}(z_i) \parallel f_i(z_i)]$, $\tilde{f}_i(z_i) = 1$ for $i > N$

repeat
for $i = 1, \dots, N$ do

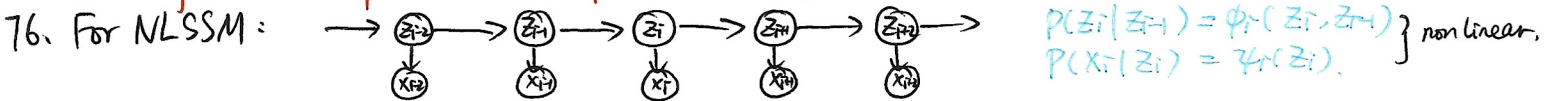
$$\text{Delete: } q_{T_i}(z) = \frac{q(z)}{\tilde{f}_i(z_i)} = \prod_{j \neq i} \tilde{f}_j(z_j)$$

$$\text{Project: } \tilde{f}_i^{\text{new}}(z_i) = \arg \min_{\tilde{f}} KL[\tilde{f}_i(z_i) q_{T_i}(z_i) \parallel f_i(z_i) q_{T_i}(z_i)]$$

$$\text{Include: } q(z) \leftarrow \tilde{f}_i^{\text{new}}(z_i) q_{T_i}(z)$$

and for
until convergence.

75. $q_{T_i}(z_i) = \prod_{j \in \text{neigh}(i)} M_{j \rightarrow i}(z_j \cap z_i)$. for trees as conditional on z_i , the rest are sub-trees and they are disjoint $\Rightarrow$ independent $\Rightarrow$ use product. For loopy case this is only an approximation.



$$f_i(z_i, z_{i-1}) = \phi_i(z_i, z_{i-1}) \psi_i(z_i) \rightarrow \text{not tractable inference.}$$

Assume $\tilde{f}_i(z_i, z_{i-1})$ is Gaussian, then $q_{T_i}(z_i, z_{i-1}) = \int \prod_{j \neq i} \tilde{f}_j(z_j, z_{j-1}) = \int \prod_{j \neq i} \tilde{f}_j(z_j, z_{j-1}) \prod_{j \neq i} \tilde{f}_j(z_j, z_j) \alpha_{i-1}(z_{i-1}) \beta_i(z_i)$

$$\tilde{f}_i(z_i, z_{i-1}) = \arg \min_{\tilde{f} \in \mathcal{F}} KL[\phi_i(z_i, z_{i-1}) \psi_i(z_i) \alpha_{i-1}(z_{i-1}) \beta_i(z_i) \parallel f_i(z_i, z_{i-1}) \alpha_{i-1}(z_{i-1}) \beta_i(z_i)]$$

$$\tilde{p}_i(z_i, z_{i-1}) = \arg \min_{p \in \mathcal{P}} KL[\tilde{p}(z_{i-1}, z_i) \parallel p(z_{i-1}, z_i)] \quad \tilde{f}_i(z_i, z_{i-1}) = \frac{\tilde{p}_i(z_i, z_{i-1})}{\alpha_{i-1}(z_{i-1}) \beta_i(z_i)}$$

$$\alpha_i(z_i) = \int \prod_{j \neq i} \tilde{f}_j(z_j, z_{j-1}) = \int \alpha_{i-1}(z_{i-1}) \tilde{f}_i(z_i, z_{i-1}) = \frac{1}{\beta_i(z_i)} \int \tilde{p}(z_{i-1}, z_i)$$

$$\beta_i(z_{i-1}) = \int \prod_{j \neq i} \tilde{f}_j(z_j, z_{j-1}) = \int \beta_i(z_i) \tilde{f}_i(z_i, z_{i-1}) = \frac{1}{\alpha_{i-1}(z_{i-1})} \int \tilde{p}(z_{i-1}, z_i)$$

77. If both $q_{T_i}(z_i)$ and $\tilde{f}$ are in the same exponential family. Let $q(x) = \frac{1}{Z(\theta)} e^{T(x) \cdot \theta}$

$$\arg \min_{\theta} KL[p(x) \parallel q(x)] = \arg \min_{\theta} KL[p(x) \parallel \frac{1}{Z(\theta)} e^{T(x) \cdot \theta}] = \arg \min_{\theta} - \int p(x) \log \frac{1}{Z(\theta)} e^{T(x) \cdot \theta} dx$$

$$= \arg \min_{\theta} - \int p(x) T(x) \cdot \theta dx + \log Z(\theta)$$

$$\frac{\partial}{\partial \theta} = - \int dx p(x) T(x) + \frac{1}{Z(\theta)} \frac{\partial}{\partial \theta} \int e^{T(x) \cdot \theta} dx = - \langle T(x) \rangle_p + \langle T(x) \rangle_q \quad \text{moment matching.}$$

78. Normal Gaussian process, $Y|g, X \sim \mathcal{N}(f(g, X), \sigma^2 I)$

$$p(g_1, \dots, g_n, y_1, \dots, y_n) = \underbrace{\mathcal{N}(g_1, \dots, g_n | 0, K)}_{\text{prior of } g, f_0(G)} \prod_i \underbrace{\mathcal{N}(y_i | g_i, \sigma_i^2)}_{f_i(g_i)}$$

If $f_i(g_i)$ is not gaussian, approximate by Gaussian $f_i(g_i) \propto \mathcal{N}(\tilde{\mu}_i, \tilde{\Sigma}_i)$

$$q(g) \propto f_0(G) \prod_i f_i(g_i) \quad \tilde{\mu} \sim \mathcal{N}(0, K + \text{diag}[\tilde{\Sigma}_i^2]) \quad q_{Ti}(g) = \frac{q(g)}{f_i(g_i)}$$

(approx posterior)

$$\begin{bmatrix} \tilde{\mu}_i \\ g_i \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \Sigma_{Ti, Ti} & \Sigma_{Ti, i} \\ \Sigma_{i, Ti} & K_{ii} \end{bmatrix}\right) \quad \text{(we can also write } \Sigma_{Ti, i} \text{ as } K_{Ti, i} \text{ as they are equal. formally should be } \text{cov}(\tilde{\mu}_i, g_i) = \text{cov}(g_{Ti} + \epsilon_{Ti}, g_i) = K_{Ti, i})$$

$$\text{so } q_{Ti}(g_i) \equiv P(g_i | \tilde{\mu}_i) = \mathcal{N}(\Sigma_{Ti, Ti}^{-1} \Sigma_{Ti, i} \tilde{\mu}_i, K_{ii} - \Sigma_{Ti, Ti}^{-1} \Sigma_{Ti, i} \Sigma_{i, Ti} \Sigma_{Ti, i}^{-1})$$

$$\text{so EP: } \arg \min_{f(g_i)} KLE[f_i(g_i) q_{Ti}(g_i) || f(g_i) q_{Ti}(g_i)] \quad \text{moment matching} \Rightarrow \tilde{f}_i^{new}(g_i) = \frac{\mathcal{N}(f_i(g_i) q_{Ti}(g_i) g_i, \text{var})}{q_{Ti}(g_i)}$$

$$\text{var} = \int f_i(g_i) q_{Ti}(g_i) g_i^2 dg_i - (\tilde{\mu}^{new})^2$$

$$\text{prediction: } P(y'|x', D) = \int \underbrace{P(y'|g')}_{\text{use true emission}} \underbrace{\mathcal{N}(g' | K_{xx'}(K_{xx} + \text{diag}[\tilde{\Sigma}_i^2])^{-1} \tilde{\mu}, K_{xx'} - K_{xx'}(K_{xx} + \text{diag}[\tilde{\Sigma}_i^2])^{-1} K_{xx'})}_{\text{approx posterior}} dg'$$

79. If $f_i(z_i) \propto e^{T(z_i) \cdot \theta_i - \phi(\theta_i)}$ (sites are exponential family), the $q(z) \propto \prod_i f_i \propto e^{T(z) \Sigma \theta_i - \Sigma \phi(\theta_i)}$.
normalise to get $q(z) = e^{T(z) \Sigma \theta_i - \phi(\Sigma \theta_i)}$.

80. Unnormalized ExpFam approximate sites $\tilde{f}_i = \tilde{C}_i e^{T(z_i) \cdot \theta_i}$, $\theta = \Sigma \theta_i$ for natural param of $q(z)$.
 $\theta_i = \Sigma_j \theta_j$ for $q_i(z)$. $\Phi(\theta) = \log \int e^{T(z) \cdot \theta} dz$, log normaliser.

81. Unnormalized KL: $KL[P || Q] = \int p(x) \log \frac{p(x)}{q(x)} dx + \frac{\int (q(x) - p(x)) \log q(x)}{\int (q(x) - p(x)) dx}$ match zero moment as well.

$$82. \int \tilde{C}_i e^{T(z_i) \cdot \theta_i} \prod_j \tilde{C}_j e^{T(z_j) \cdot \theta_j} = \int f_i(z_i) \prod_j \tilde{C}_j e^{T(z_j) \cdot \theta_j} \Rightarrow \tilde{C}_i = e^{\hat{\Phi}_i(\theta_i) - \Phi(\theta)}$$

$$\frac{\int f_i(z_i) \prod_j e^{T(z_j) \cdot \theta_j} dz}{\int f_i(z_i) e^{T(z_i) \cdot \theta_i} dz} = e^{\hat{\Phi}_i(\theta_i)} \quad \hat{\Phi}_i(\theta_i) = \log \text{normaliser of tilted ExpFam } \tilde{P}_i(z_i) \propto f_i(z_i) e^{T(z_i) \cdot \theta_i}$$

$$83. \log \int \prod_i f_i(z_i) \approx \log \int \prod_i \tilde{f}_i(z_i) = \log(\prod_i \tilde{C}_i \int e^{T(z) \cdot \theta}) = \Phi(\theta) + \sum \log \tilde{C}_i =: \tilde{\mathcal{L}}$$

Earlier we have written $P(Z|X) = \frac{1}{Z} \prod_i f_i(z_i) = \frac{1}{P(X)} \prod_i f_i(z_i)$ so $P(X) = \int \prod_i f_i(z_i) dz$
so $\tilde{\mathcal{L}}$ is approximate log likelihood.

84. Let f_i depend on model (hyper)parameter θ .

$$\begin{aligned} \nabla_{\theta} \tilde{\mathcal{L}} &= \nabla_{\theta} \Phi(\theta) + \sum_i \nabla_{\theta} \log \tilde{C}_i = \mu \nabla_{\theta} \theta + \sum_i \nabla_{\theta} \log \tilde{C}_i \quad \mu = \nabla_{\theta} \Phi(\theta) \\ &= \mu \nabla_{\theta} \theta + \sum_i (\nabla_{\theta} \hat{\Phi}_i(\theta_i) - \mu \nabla_{\theta} \theta) \quad \hat{\Phi}_i(\theta_i) = \log \int f_i(z_i) e^{T(z_i) \cdot \theta_i} \\ &= \mu \nabla_{\theta} \theta + \sum_i (\mu \cdot \nabla_{\theta} \theta_i + \langle \nabla_{\theta} \log f_i(z_i) \rangle_{\tilde{P}_i} - \mu \nabla_{\theta} \theta) \\ &= \mu \nabla_{\theta} (\theta + \sum_i (\theta_i - \theta)) + \sum_i \langle \nabla_{\theta} \log f_i(z_i) \rangle_{\tilde{P}_i} \\ &= \sum_i \langle \nabla_{\theta} \log f_i(z_i) \rangle_{\tilde{P}_i} \end{aligned}$$

$$\begin{aligned} \nabla_{\theta} \hat{\Phi}_i(\theta_i) &= \nabla_{\theta} \log \int f_i(z_i) e^{T(z_i) \cdot \theta_i} = \underbrace{\partial_{\theta_i} \hat{\Phi}_i(\theta_i) \cdot \nabla_{\theta} \theta_i}_{\text{through } \theta_i} + \underbrace{e^{-\hat{\Phi}_i(\theta_i)} \int \nabla_{\theta} f_i(z_i) e^{T(z_i) \cdot \theta_i} dz}_{\text{through } f_i(z_i)} \\ &= \langle T(z) \rangle_{\tilde{P}_i} \cdot \nabla_{\theta} \theta_i + \int \nabla_{\theta} \log f_i(z_i) f_i(z_i) e^{T(z_i) \cdot \theta_i - \hat{\Phi}_i(\theta_i)} dz \\ &= \mu \cdot \nabla_{\theta} \theta_i + \langle \nabla_{\theta} \log f_i(z_i) \rangle_{\tilde{P}_i} \quad \langle T(z) \rangle_{\tilde{P}_i} = \mu \text{ in EP} \end{aligned}$$

85. Alpha divergence: $D_{\alpha}[P || Q] = \frac{1}{\alpha(1-\alpha)} \int (\alpha p(x) + (1-\alpha)q(x) - p(x)^{\alpha} q(x)^{1-\alpha}) dx$

$$\lim_{\alpha \rightarrow 0} D_{\alpha}[P || Q] = KL[P || Q] \quad \text{as } \lim_{\alpha \rightarrow 0} \frac{(p(x)/q(x))^{\alpha}}{\alpha} = \log \frac{p(x)}{q(x)}$$

$$\lim_{\alpha \rightarrow 1} D_{\alpha}[P || Q] = KL[P || Q] \quad \tilde{f}_i^{new} = \arg \min_{f \in \mathcal{F}} KLE[f_i(z_i)^{\alpha} \tilde{f}_i^{\alpha}(z_i)^{1-\alpha} q_{Ti}(z_i) || f(z_i) q_{Ti}(z_i)]$$

$\alpha < 1$, we have stable results (small update from $\tilde{f}_i^{\text{old}}(z_i)$)

86. Deal with loops:

- Damping: $M_{i \rightarrow j}^{\text{new}}(X_j) = (1-\alpha) M_{i \rightarrow j}^{\text{old}}(X_j) + \alpha \left(\sum_{X_i} f_i(X_i) f_j(X_j) \prod_{k \in \text{ne}(j) \setminus i} M_{k \rightarrow j}(X_j) \right) M_{i \rightarrow j}(X_i)$

- Grouping variable into cliques to improve accuracy.

87. Loopy BP as message-based expectation propagation: $(X_i) \square (X_j) \Rightarrow (X_i) \square \square (X_j)$

Approximate pairwise factors f_{ij} by product of messages $f_{ij}(X_i, X_j) \approx \tilde{f}_{ij}(X_i, X_j) = M_{i \rightarrow j}(X_j) M_{j \rightarrow i}(X_i)$.

Full joint is approximated as $P(X) \approx \frac{1}{Z} \prod_{\text{nodes } i} f_i(X_i) \prod_{\text{edges } (i,j)} \tilde{f}_{ij}(X_i, X_j) = \frac{1}{Z} \prod_{\text{nodes } i} \left(f_i(X_i) \prod_{j \in \text{ne}(i)} M_{j \rightarrow i}(X_i) \right)$

$$b_i(X_i) \propto f_i(X_i) \prod_{j \in \text{ne}(i)} M_{j \rightarrow i}(X_i)$$

EP: Deletion: $q_{ij}(X) = f_i(X_i) f_j(X_j) \prod_{k \in \text{ne}(i) \setminus j} M_{k \rightarrow i}(X_i) \prod_{l \in \text{ne}(j) \setminus i} M_{l \rightarrow j}(X_j) \prod_{s \neq i,j} f_s(X_s) \prod_{t \in \text{ne}(s)} M_{t \rightarrow s}(X_s)$

$$q_{ij}(X_i, X_j) = f_i(X_i) f_j(X_j) \prod_{k \in \text{ne}(i) \setminus j} M_{k \rightarrow i}(X_i) \prod_{l \in \text{ne}(j) \setminus i} M_{l \rightarrow j}(X_j)$$

Projection: $\tilde{f}_{ij}(X_i, X_j) = \{M_{i \rightarrow j}^{\text{new}}, M_{j \rightarrow i}^{\text{new}}\} = \arg \min_{KL} [f_{ij}(X_i, X_j) q_{ij}(X_i, X_j) \parallel M_{i \rightarrow j}(X_j) M_{j \rightarrow i}(X_i) q_{ij}(X_i, X_j)]$

$$q_{ij}(X_i) = f_i(X_i) \prod_{k \in \text{ne}(i) \setminus j} M_{k \rightarrow i}(X_i) \quad q_{ij}(X_j) = f_j(X_j) \prod_{l \in \text{ne}(j) \setminus i} M_{l \rightarrow j}(X_j)$$

$$q_{ij}(X_i, X_j) = q_{ij}(X_i) q_{ij}(X_j)$$

$$b_i(X_i) \propto M_{j \rightarrow i}^{\text{new}}(X_i) q_{ij}(X_i) \propto \sum_{X_j} (f_j(X_j) \prod_{k \in \text{ne}(j) \setminus i} M_{k \rightarrow j}(X_j)) f_i(X_i) \prod_{l \in \text{ne}(i) \setminus j} M_{l \rightarrow i}(X_i)$$

$$\text{so } M_{j \rightarrow i}^{\text{new}}(X_i) = \sum_{X_j} (f_j(X_j) \prod_{k \in \text{ne}(j) \setminus i} M_{k \rightarrow j}(X_j))$$

note in tree case $b_i(X_i) \propto P(X_i)$ so it is marginal

Message based EP reexpressed in message $M_{i \rightarrow j}(X_j)$ instead of sites $f_i(X_i)$ so we don't need to approximate both site and cavity. $\tilde{f}_i(X_i) \approx \prod_{j \in \text{ne}(i)} M_{j \rightarrow i}(X_i \mid Z_j \setminus X_i)$. (a product of exp fam).

88. Loopy BP as tree-based reparameterization.

$$P(X) = \frac{1}{Z} \prod_{\text{nodes } i} f_i(X_i) \prod_{\text{edges } (i,j)} f_{ij}(X_i, X_j) = \underbrace{P(X_i)}_{\text{undirected tree}} \underbrace{\prod_{i \neq j} P(X_i \mid X_{\text{pa}(i)})}_{\text{DAG}} = \prod_{\text{nodes } i} \underbrace{P(X_i) \prod_{j \in \text{ne}(i)} \frac{P(X_i, X_j)}{P(X_i) P(X_j)}}_{\text{pairwise marginal, } \sum_{X_j} P(X_i, X_j) = P(X_i)}$$

From $P(X) \propto \prod_{(i,j)} f_{ij}(X_i, X_j)$ to $P(X) = \prod_i P(X_i) \prod_{(i,j)} \frac{P(X_i, X_j)}{P(X_i) P(X_j)}$

Define $\tilde{f}_{ij} = f_{ij}$, $\tilde{f}_i = P_i = 1$. Iterate over edges.

$$P^n(X_i, X_j) = \frac{1}{Z_{ij}^n} \tilde{f}_i^{n-1}(X_i) \tilde{f}_{ij}^{n-1}(X_i, X_j) \tilde{f}_j^{n-1}(X_j), \quad \boxed{\tilde{f}_i^n(X_i)} = P^n(X_i) = \sum_{X_j} P^n(X_i, X_j) = \boxed{\tilde{f}_i^{n-1}(X_i)} \sum_{X_j} \tilde{f}_{ij}^{n-1}(X_i, X_j) \tilde{f}_j^{n-1}(X_j)$$

$$\tilde{f}_{ij}^n(X_i, X_j) = \frac{\tilde{f}_{ij}^{n-1}(X_i, X_j)}{M_{j \rightarrow i}^{n-1}(X_i)}, \text{ we need to scale down by } M_{j \rightarrow i}^{n-1}(X_i) \text{ since we multiply } M_{j \rightarrow i} \text{ in } \tilde{f}_i(X_i).$$

$$\tilde{f}_i^{\infty}(X_i) = 1 \times \prod_{j \in \text{ne}(i)} M_{j \rightarrow i}(X_i) = P(X_i)$$

$$\tilde{f}_{ij}^{\infty}(X_i, X_j) = \frac{f_{ij}(X_i, X_j)}{M_{j \rightarrow i}(X_i) M_{i \rightarrow j}(X_j)} = \frac{\left(\prod_{k \in \text{ne}(i) \setminus j} M_{k \rightarrow i}(X_i) \right) f_{ij}(X_i, X_j) \left(\prod_{l \in \text{ne}(j) \setminus i} M_{l \rightarrow j}(X_j) \right)}{\left(\prod_{k \in \text{ne}(i) \setminus j} M_{k \rightarrow i}(X_i) \right) M_{j \rightarrow i}(X_i) M_{i \rightarrow j}(X_j) \left(\prod_{l \in \text{ne}(j) \setminus i} M_{l \rightarrow j}(X_j) \right)} = \frac{P(X_i, X_j)}{P(X_i) P(X_j)}$$

On a non-tree, it will successfully reparameterized the distribution to have locally consistent beliefs.

$$P(X) \propto \prod_i b(X_i) \prod_{(i,j)} \frac{b(X_i, X_j)}{b(X_i) b(X_j)} \text{ with } \sum_{X_j} b(X_i, X_j) = b(X_i)$$

marginal is not usually correct. It is not global consistent.

$$\sum_{X_i} \left(\prod_i b(X_i) \prod_{(i,j)} \frac{b(X_i, X_j)}{b(X_i) b(X_j)} \right) \neq b(X_i)$$

This pseudo marginals present valid belief for any of the spanning tree of loopy.

89. Bethe free energy defines on locally consistent beliefs:

$$\sum_{x_i} b_i(x_i) = 1, \forall i \quad \text{and} \quad \sum_{x_j} b_{ij}(x_i, x_j) = b_i(x_i) \quad \forall (i, j) \in \text{ne}(i), x_i.$$

$$\text{Variational free energy: } F(\mathbf{b}) = \langle \log P(\mathbf{X}) \rangle_{\mathbf{b}} + H(\mathbf{b})$$

$$\text{negative Bethe free energy: } F_{\text{Bethe}}(\mathbf{b}) = E_{\text{Bethe}}(\mathbf{b}) + \mathcal{H}_{\text{Bethe}}(\mathbf{b}).$$

$$P(\mathbf{X}) \propto \prod_i f_i(x_i) \prod_{(i,j)} f_{ij}(x_i, x_j) \propto \prod_i b_i(x_i) \prod_{(i,j)} \frac{b_{ij}(x_i, x_j)}{b_i(x_i) b_j(x_j)}$$

$$E_{\text{Bethe}}(\mathbf{b}) = \sum_i \sum_{x_i} b_i(x_i) \log f_i(x_i) + \sum_{(i,j)} \sum_{x_i, x_j} b_{ij}(x_i, x_j) \log f_{ij}(x_i, x_j) \quad \text{use } b_i \text{ as if it is true marginal.}$$

$$\mathcal{H}_{\text{Bethe}}(\mathbf{b}) = \sum_i H[b_i] - \sum_{(i,j)} \text{KL}[b_{ij} \| b_i b_j] = -\sum_i \sum_{x_i} b_i(x_i) \log b_i(x_i) - \sum_{(i,j)} \sum_{x_i, x_j} b_{ij}(x_i, x_j) \log \frac{b_{ij}(x_i, x_j)}{b_i(x_i) b_j(x_j)}.$$

On a tree $F_{\text{Bethe}} = F$.

90. Bethe free energy Lagrangian is

$$\mathcal{L} = E_{\text{Bethe}}(\mathbf{b}) + \mathcal{H}_{\text{Bethe}}(\mathbf{b}) + \sum_i \lambda_i \left(\sum_{x_i} b_i(x_i) - 1 \right) + \sum_{(i,j)} \left[\sum_{x_i} \lambda_{ij}(x_i) \left(\sum_{x_j} b_{ij}(x_i, x_j) - b_i(x_i) \right) + \sum_{x_j} \lambda_{ji}(x_j) \left(\sum_{x_i} b_{ij}(x_i, x_j) - b_j(x_j) \right) \right]$$

$$\frac{\partial \mathcal{L}}{\partial b_i(x_i)} = \log f_i(x_i) - \log b_i(x_i) + \sum_{j \in \text{ne}(i)} \sum_{x_j} \frac{b_{ij}(x_i, x_j)}{b_i(x_i)} + \lambda_i - \sum_{j \in \text{ne}(i)} \lambda_{ij}(x_i) = 0.$$

$$\Rightarrow b_i(x_i) \propto f_i(x_i) \prod_{j \in \text{ne}(i)} e^{-\lambda_{ij}(x_i)}$$

$$\frac{\partial \mathcal{L}}{\partial b_{ij}(x_i, x_j)} = \log f_{ij}(x_i, x_j) - \log b_{ij}(x_i, x_j) + \log b_i(x_i) b_j(x_j) + \lambda_{ij}(x_i) + \lambda_{ji}(x_j) = 0$$

$$\Rightarrow b_{ij}(x_i, x_j) \propto f_{ij}(x_i, x_j) b_i(x_i) b_j(x_j) e^{\lambda_{ij}(x_i)} e^{\lambda_{ji}(x_j)}.$$

Compare to BP $M_{j \rightarrow i}(x_i) = e^{-\lambda_{ij}(x_i)}.$

solving for $\lambda_{ij}(x_i)$ by enforcing constraint $\sum_{x_j} b_{ij}(x_i, x_j) = b_i(x_i)$, we have.

$$\sum_{x_j} b_{ij}(x_i, x_j) \propto \sum_{x_j} f_{ij}(x_i, x_j) b_i(x_i) b_j(x_j) e^{\lambda_{ij}(x_i)} e^{\lambda_{ji}(x_j)}$$

$$\Rightarrow b_i(x_i) \propto b_i(x_i) e^{\lambda_{ij}(x_i)} \sum_{x_j} f_{ij}(x_i, x_j) b_j(x_j) e^{\lambda_{ji}(x_j)}.$$

$$\Rightarrow e^{-\lambda_{ij}(x_i)} \propto \sum_{x_j} f_{ij}(x_i, x_j) \frac{b_j(x_j) e^{\lambda_{ji}(x_j)}}{b_i(x_i)} = \sum_{x_j} f_{ij}(x_i, x_j) f_j(x_j) \prod_{(j,k) \in \text{ne}(j)} e^{-\lambda_{jk}(x_j)}.$$

BP message passing rule.

91. Fixed point of loopy BP are exactly the stationary points of the Bethe free energy.

92. Bethe free energy is an approximation based on inclusion-exclusion until second order.

$$\mathcal{H}_{\text{Bethe}}(\mathbf{b}) = \sum_i H[b_i] - \sum_{(i,j)} \text{KL}[b_{ij} \| b_i b_j].$$

93. Bethe free energy accounts for interactions between different sites, while variational free energy assumes independence.

94. $\Phi(\theta)$ is log partition function in expfam $P(\mathbf{X}|\theta) = \exp(\theta^T S(\mathbf{X}) - \Phi(\theta)).$

$$\frac{\partial \Phi(\theta)}{\partial \theta} = \mu \quad \frac{\partial^2 \Phi(\theta)}{\partial \theta^2} = \text{Var}(S(\mathbf{X})) \quad \text{so } \Phi(\theta) \text{ is convex in } \theta$$

95. Negative Entropy: $\psi(\mu) = \mathbb{E}_{\theta}[\log P(\mathbf{X}|\theta(\mu))] = \mathbb{E}_{\theta}[\theta^T S(\mathbf{X}) - \Phi(\theta)] = \theta^T \mu - \Phi(\theta).$

$$\text{so } \theta^T \mu = \Phi(\theta) + \psi(\mu).$$

96. Negative Entropy is dual to the log-partition function.

97. Legendre-Fenchel conjugate: $f^*(y) = \sup_{x \in \mathbb{R}^n} (\langle y, x \rangle - f(x))$

f^* is always convex. $\langle y, x \rangle \leq f(x) + f^*(y) \quad \forall x, y.$

98. Fenchel-Young inequality: $f(x) + f^*(y) = x^T y \iff y \in \partial f(x)$

$$\text{so } \mu = \nabla \Phi(\theta) \quad \theta = \nabla \psi(\mu).$$

99. $KL[\theta || \theta'] = KL[P(X|\theta) || P(X|\theta')] = \mathbb{E}_\theta[-\log P(X|\theta')] + \log P(X|\theta) = -\theta'^T \mu + \xi(\theta') + \psi(\mu) \geq 0$

So. $\psi(\mu) = \sup_{\theta'} [\theta'^T \mu - \xi(\theta')]$ and continuous functions are reciprocally dual, so

$$\xi(\theta) = \sup_{\mu'} [\theta^T \mu' - \psi(\mu')]$$

100. $p(x, z) = \exp(\theta^T S(x, z) - \xi_{xz}(\theta))$. Posterior on z is also ExpFam, with clamped sufficient statistic

$S_z(z|x) = S_{xz}(x^{obs}, z)$. Same natural parameter θ and $\xi_z(\theta) = \log \sum_z \exp \theta^T S_z(z)$

$$L(\theta) = p(x|\theta) = \sum_z e^{\theta^T S(x, z) - \xi_{xz}(\theta)} = \sum_z e^{\theta^T S_z(z|x) - \xi_{xz}(\theta)} = \exp(\xi_z(\theta) - \xi_{xz}(\theta)).$$

$$\ell(\theta) = \sup_{\mu_z} [\theta^T \mu_z - \psi(\mu_z) - \xi_{xz}(\theta)] = \sup_{\mu_z} [\underbrace{\theta^T \mu_z - \xi_{xz}(\theta)}_{\langle \log(x, z) \rangle_\theta} - \underbrace{\psi(\mu_z)}_{-H(q)}] = \sup_{\mu_z} F(\theta, \mu_z) \quad (q_z \text{ represented by } \mu_z).$$

$$\mu_z^* = \operatorname{argmax}_{\mu_z} [\theta^T \mu_z - \psi(\mu_z)] \quad \text{feasible set: } \mu = \sum_x \nu(x) S(x) \quad \sum_x \nu(x) = 1$$

101. Discrete MRF: $P(X) = \frac{1}{Z} \prod_i f_i(x_i) \prod_{(i,j)} f_{ij}(x_i, x_j) = \exp\left(\sum_i \sum_k \theta_i(k) \delta(x_i=k) + \sum_{i,j} \sum_{k,l} \theta_{ij}(k,l) \delta(x_i=k) \delta(x_j=l) - \xi(\theta)\right)$

natural parameters: $\theta = [\theta_i(k), \theta_{ij}(k,l)] \cdot \forall i, j, k, l$

mean of sufficient statistics: $\mu = [p(X_i=k), p(X_i=k, X_j=l)] \cdot \forall i, j, k, l$.

If MRF has tree structure:

$$\psi(\mu_T) = \mathbb{E}_\theta \left[\log \prod_i P(X_i) \prod_{(i,j)} \frac{P(X_i, X_j)}{P(X_i)P(X_j)} \right] = -\sum_i \underbrace{H(X_i)}_{\text{Entropy}} + \sum_{(i,j)} \underbrace{I(X_i, X_j)}_{\text{mutual information}}.$$

102. $\psi_{\text{Bethe}}(\mu_z) = -\sum_i H(X_i) + \sum_{(i,j) \in G} I(X_i, X_j)$ is not convex!

103. Imagine a set of spanning trees T for MRF, each with its own parameters θ_T, μ_T . By padding entries corresponding to off-tree edges with zero, we can assume θ_T has same dimension as θ .

Suppose we have a distribution β on spanning trees so that $\mathbb{E}_\beta[\theta_T] = \theta$.

Then $\xi(\theta) = \xi(\mathbb{E}_\beta[\theta_T]) \leq \mathbb{E}_\beta[\xi(\theta_T)]$

We can tighten the bound to get a good approximation to ξ : $\xi(\theta) \leq \sup_{\beta, \theta_T: \mathbb{E}_\beta[\theta_T] = \theta} \mathbb{E}_\beta[\xi(\theta_T)] =: \xi_\beta(\theta)$.

Lagrange: $\mathcal{L} = \mathbb{E}_\beta[\xi(\theta_T)] - \lambda^T (\mathbb{E}_\beta[\theta_T] - \theta)$.

$\frac{\partial \mathcal{L}}{\partial \theta_T} = 0 \Rightarrow \mu_T = \Pi_T(\lambda)$, $\Pi_T(\lambda)$ selects multipliers corresponding to elements of θ that are non-zero in the tree T .

$$\begin{aligned} \text{dual: } \xi_\beta(\theta) &= \sup_{\theta_T} [\inf_{\beta} \mathbb{E}_\beta[\xi(\theta_T)] - \lambda^T (\mathbb{E}_\beta[\theta_T] - \theta)] \\ &= \sup_{\lambda} [\lambda^T \theta + \mathbb{E}_\beta[\inf_{\theta_T} \xi(\theta_T) - \theta_T^T \Pi_T(\lambda)]] \\ &= \sup_{\lambda} [\lambda^T \theta + \mathbb{E}_\beta[-\psi(\Pi_T(\lambda))]] \\ &= \sup_{\lambda} [\lambda^T \theta + \mathbb{E}_\beta[\sum_i H(X_i) - \sum_{(i,j) \in T} I(X_i, X_j)]] \\ &= \sup_{\lambda} [\lambda^T \theta + \sum_i H(X_i) - \sum_{(i,j)} \beta_{ij} I(X_i, X_j)] \quad \text{singleton is always selected in } T. \end{aligned}$$

Convexified version of Bethe free energy.

104. REINFORCE trick: $\nabla_\theta F(p, \theta) = \nabla_p \int q(z; p) (\log P(x, z|\theta) - \log q(z; p)) dz$
 $= \int \nabla_p q(z; p) (\log P(x, z|\theta) - \log q(z; p)) dz$
 $+ \int q(z; p) \nabla_p (\log P(x, z|\theta) - \log q(z; p)) dz$
 $= \int \nabla_p \log q(z; p) (\log P(x, z|\theta) - \log q(z; p)) q(z; p) dz$
 $= \langle \nabla_p \log q(z; p) (\log P(x, z|\theta) - \log q(z; p)) \rangle_{q(z; p)}.$

105. Recognition model and amortized inference:

- $\{x_i, z\}$ z is still latent
- learn $q(z_i; p = f(x_i, \phi))$
- Optimal variational distribution $q^*(z)$ depends on x_i .
- f is recognition model and this approach is called amortized inference. we learn $f(x_i, \phi)$ instead of q .

106. Wake-sleep learning:

phase Wake: Generative model $p_\theta(x, z)$, with real data x .

- infer $\hat{z} = f(x; \phi)$ with current f
- update generative parameter θ . $\theta \leftarrow \theta + \eta \nabla_\theta \log p_\theta(x, \hat{z})$.

phase sleep: Recognition model $q(z, \phi = f(x, \phi))$, sample $\{z_s, x_s\}$ from $p_\theta(x, z)$

- update to minimise $KL[p_\theta(z|x) || q(z; f(x, \phi))] \Rightarrow \min -\log q(z; f(x, \phi))$.
- $\Delta \phi \propto \sum_s (z_s - f(x_s; \phi)) \nabla_\phi f(x_s; \phi)$

Wake: freeze ϕ , update θ to explain real data.

Sleep: freeze θ , update ϕ to recover latents on model generated data.

107. Variational Autoencoder: $P(z) = \mathcal{N}(0, I)$ $P(x|z) = \mathcal{N}(g_{NN}(z; \theta), \sigma^2 I)$

$g_{NN}(z; \theta)$ is Gaussian with neural network for θ .

posterior approximation (encoder): $q(z|x; f(x, \phi))$

$$F(\theta, \phi) = \langle \sum_{\text{data}} \log P(x|z) + \log P(z) \rangle_{q(z|x)} + H[q(z|x)] = \sum_{\text{data}} \langle \log P(x|z) \rangle_{q(z|x)} - KL[q(z|x) || P(z)]$$

$$= - \sum_{\text{data}} \underbrace{\langle \frac{\|x - g(z)\|^2}{2\sigma^2} \rangle_q}_{\text{reconstruction cost}} - \underbrace{KL[q(z|x) || P(z)]}_{\text{regulariser}}$$

108. Reparametrisation trick: If f gives marginal with μ_i and σ_i for z_i , we can use $z_i^s = \mu_i + \sigma_i \cdot \epsilon_i^s$

$\epsilon_i^s \sim \mathcal{N}(0, 1)$.

$$F_i(\theta, \phi) = \frac{1}{S} \sum_s \log P(x_i, z_i^s; \theta) - \log q(z_i^s; f(x_i, \phi))$$

$$\frac{\partial F_i}{\partial \theta} = \frac{1}{S} \sum_s \nabla_\theta \log P(x_i, z_i^s; \theta) \quad \frac{\partial F_i}{\partial \phi} = \frac{1}{S} \sum_s \frac{\partial}{\partial z_i^s} (\log P(x_i, z_i^s; \theta) - \log q(z_i^s; f(x_i, \phi))) \frac{\partial z_i^s}{\partial \phi}$$

$$+ \frac{\partial}{\partial f(x_i)} \log q(z_i^s; f(x_i)) \frac{df(x_i)}{d\phi}$$

Train θ and ϕ together, holding ϵ^s fixed.

109. Importance-weighted free energy:

$$\ell(\theta) = \log P(x) = \log \mathbb{E}_{z \sim p(z|x)} [P(x)] = \log \mathbb{E}_{z \sim q} \left[\frac{P(x)P(z|x)}{q(z)} \right] \geq \mathbb{E}_{z \sim q} \left[\log \frac{P(x)P(z|x)}{q(z)} \right] = F(q, \theta)$$

$$\ell(\theta) = \log \mathbb{E}_{z_1, \dots, z_K \sim q} \left[\frac{1}{K} \sum_k \frac{P(x, z_k)}{q(z_k)} \right] \geq \mathbb{E}_{z_1, \dots, z_K \sim q} \left[\log \frac{1}{K} \sum_k \frac{P(x, z_k)}{q(z_k)} \right]$$

$K \rightarrow \infty$, the signal for learning amortised q vanish as we lose the gradient of q

$$\nabla_\phi L_K = \mathbb{E}_{z_1, \dots, z_K \sim q} \left[\sum_{k=1}^K \left(\frac{w_k}{\sum_{j=1}^K w_j} \right) \nabla_\phi \log q(z_k) \right] \quad w_k = \frac{P(x, z_k)}{q(z_k)} \quad \text{large } K: \frac{w_k}{\sum w_j} \approx \frac{1}{K}$$

$$\text{and } \nabla_\phi L_K \approx \frac{1}{K} \sum_{k=1}^K \nabla_\phi \log q(z_k) \rightarrow 0 \text{ as } \mathbb{E}[\nabla_\phi \log q(z)] = 0$$

small K : w_k dominates at some k , gradient is strong and $q(z)$ learning focus on high-likelihood regions.

110. Normalising flows:

Consider a recognition model $q(z)$ implicitly by $z_0 \sim q_0(\cdot; x) \leftarrow$ a fixed, tractable one, eg. $\mathcal{N}(x, I)$.

then $z = f_K(f_{K-1}(\dots f_1(z_0)))$, f_k is a smooth, invertible parametrised by ϕ .

$$\langle F(z) \rangle_q = \langle F(f_K(f_{K-1}(\dots f_1(z_0)))) \rangle_{q_0} \text{ and } \log q(z) = \log q_0(f_1^{-1}(f_2^{-1}(\dots f_K^{-1}(z)))) - \sum_k \log |\nabla f_k|$$

$$\text{since we have } z_k = f_k(z_{k-1}) \Rightarrow q(z_{k-1}) = q(f_k^{-1}(z_k)) \left| \frac{\partial z_{k-1}}{\partial z_k} \right| = q(f_k^{-1}(z_k)) |\nabla f_k(z_{k-1})|^{-1}$$

Given a sample $z_0^s \sim q_0(\cdot; x)$:

$$F(\phi, \theta) \approx \frac{1}{S} \sum_s \log P(x, f_K(\dots f_1(z_0^s))) + H[q_0] + \frac{1}{S} \sum_s \sum_k \log |\nabla f_k(f_{k-1}(\dots f_1(z_0^s)))|$$

11. Diffusion model:

- Fixed "diffusion" recognition model ("forward" model)

$$q(z_1|x) = \mathcal{N}(\sqrt{1-\beta_1} x, \beta_1 I)$$

$$q(z_k|z_{k-1}) = \mathcal{N}(\sqrt{1-\beta_k} z_{k-1}, \beta_k I). \quad q(z_k) \xrightarrow{k \rightarrow \infty} \mathcal{N}(0, I).$$

- parametrize generative model ("backward" model)

$$p(z_k) = \mathcal{N}(0, I)$$

$$p(z_{k+1}|z_k; \theta) = \mathcal{N}(\mu_\theta(z_k, k), \Sigma_\theta(z_k, k)).$$

$$p(x|z_1; \theta) = \mathcal{N}(\underbrace{\mu_\theta(z_1, 1)}_{\text{usually NMs}}, \Sigma_\theta(z_1, 1))$$



$$F = \langle \log p(x|z_1) + \sum_{k=2}^K \log p(z_{k+1}|z_k) + \log p(z_k) \rangle_{q(z_{1:K}|x)} - H[q(z_{1:K}|x)]$$

$$= \langle \log p(x|z_1) \rangle_{q(z_1|x)} + \sum_{k=2}^K \langle \log p(z_{k+1}|z_k) \rangle_{q(z_k, z_{k+1}|x)} + \langle \log p(z_k) \rangle_{q(z_k|x)} - \sum_{k=1}^K H[\cdot]$$

$$q(z_1|x) = \mathcal{N}(\sqrt{1-\beta_1} x, \beta_1 I) \quad q(z_2|x) = \mathcal{N}(\sqrt{1-\beta_2} \sqrt{1-\beta_1} x, \sqrt{1-\beta_2} (\beta_1 I) \sqrt{1-\beta_2} + \beta_2 I) = \mathcal{N}(\sqrt{1-\beta_2} \sqrt{1-\beta_1} x, (1-\beta_2)(1-\beta_1) I)$$

$$\text{let } \alpha_k = 1 - \beta_k \text{ and } \bar{\alpha}_k = \prod_{i=1}^k \alpha_i \text{ and suppose } q(z_k|x) = \mathcal{N}(\sqrt{\bar{\alpha}_k} x, (1 - \bar{\alpha}_k) I)$$

$$\Rightarrow q(z_{k+1}|x) = \mathcal{N}(\sqrt{\bar{\alpha}_{k+1}} \sqrt{\bar{\alpha}_k} x, \bar{\alpha}_{k+1} (1 - \bar{\alpha}_k) I + \beta_{k+1} I) = \mathcal{N}(\sqrt{\bar{\alpha}_{k+1}} x, (1 - \bar{\alpha}_{k+1}) I).$$

11.2. Recognition-parametrized model (RPM):

$$p_{\theta, x^n}(\chi, z) = \underbrace{p(z)}_{\text{prior}} \prod_j \underbrace{p_0(x_j)}_{\text{just data}} \frac{f_{\theta_j}(z|x_j)}{F_{\theta_j}(z)}$$

$$p_0(x_j) = \frac{1}{N} \sum \delta(x_j - x_j^{(n)}).$$

$f_{\theta_j}(z|x_j)$ parametrized recognition factor, non-linear and conjugate to $p(z)$

$$F_{\theta_j}(z) = \int p_0(x_j) f_{\theta_j}(z|x_j) dx_j = \frac{1}{N} \sum f_{\theta_j}(z|x_j^{(n)}).$$